State-Dependent Uncertainty Bounds for Robust Output-Feedback MPC

Derivation and a Falcon Certificate-Reuse Numerical Example

Research draft

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I Introduction

Robust output-feedback MPC must account for both uncertain state estimates and deviations between an observer trajectory and a nominal prediction. A global uncertainty envelope can be unnecessarily conservative in operating regions with smaller model mismatch. Our objective is to retain an existing offline design while allowing the predicted uncertainty allowance to depend on the operating region covered by the error tubes.

The organization follows [1]: preliminaries, estimation bounds, robust output-feedback MPC, and a numerical example. Joint nominal and tube prediction, a free nominal initial center, and shifted-candidate arguments are established ingredients of Section IV-B of that reference. They are not claimed as new here. Our proposed extension concerns the local process-envelope construction and its incorporation into coupled continuous-time radius dynamics.

We consider robust output-feedback MPC with fixed metrics and observer/controller gains, a fixed regulation target, and time-invariant constraints.

II Preliminaries

A Plant, observer, and nominal prediction

Consider

$$\dot{x} = f_0(x,u) + E_0 w_x, \qquad w_x \in \mathcal{W}(x,u), \tag{II.1a}$$

$$y = Cx + F\eta, \qquad \eta \in \mathcal{N}, \tag{II.1b}$$

$$\dot{\hat{x}} = f_0(\hat{x},u) + L(y - C\hat{x}), \tag{II.1c}$$

$$\dot{z} = f_0(z,v), \qquad u = v + K(\hat{x} - z). \tag{II.1d}$$

Here $x, \hat{x}, z \in \mathbb{R}^n$ and $u, v \in \mathbb{R}^{n_u}$; v is the nominal control input, $E_0 \in \mathbb{R}^{n\times m}$ maps process-disturbance components into the state dynamics, and $\mathcal{W}(x,u) \subset \mathbb{R}^m$ describes their admissible values at the current state and input. The physical process disturbance is $d = E_0 w_x$. Measurement noise belongs to the fixed bounded set $\mathcal{N}$, and L is a fixed observer gain. The nominal model f_0 includes any identified constant bias, so w_x represents the centered residual uncertainty.

We assume a global baseline box encloses the process uncertainty on the operating region under consideration:

$$\mathcal{W}(x,u) \subseteq \mathcal{W}_0, \qquad \mathcal{W}_0 = \{w \in \mathbb{R}^m : |w_j| \le \bar{w}_j,\ j = 1,\dots,m\}, \quad \bar{w}_j > 0.$$

The local sets $\mathcal{W}(x,u)$ specify the state–input dependence. Their containment in $\mathcal{W}_0$ permits use of a global baseline certificate. Section III-B introduces the particular family of local sets used to obtain an explicit certificate reduction. Let $P = P_\delta \succ 0$, $X = P^{-1}$, and $T^\top T = P$. Define

$$e = x - \hat{x}, \quad \delta = \hat{x} - z, \quad V_e = e^\top P e, \quad V_e \le R, \quad \|\delta\|_P \le s. \tag{II.2}$$

R is a squared radius and $r = \sqrt{R}$ is a radius; neither denotes the true estimation error. The symbol P is distinct from a terminal-cost matrix. Since $x - z = e + \delta$,

$$\|x - z\|_P \le \sqrt{R} + s. \tag{II.3}$$

Subtracting the dynamics gives

$$\dot{e} = f_0(x, u) - f_0(\hat{x}, u) - LCe + E_0 w_x - LF\eta, \tag{II.4a}$$
$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \tag{II.4b}$$

Process disturbance enters the estimation error directly. It influences observer-to-nominal tracking indirectly through e; the observer equation itself contains no additive plant process disturbance.

B Time and prediction notation

Actual signals use a single time subscript, for example $x_t, \hat{x}_t, u_t$, with continuous physical time t. Plans are computed at sampling instants $t \in \{0, h, 2h, \ldots\}$, where $h > 0$ is the sampling period. Predicted trajectories use paired subscripts: $z_{\tau|t}, v_{\tau|t}, R_{\tau|t}, s_{\tau|t}$ denote predictions computed at time t for physical time $t + \tau$, with continuous prediction time $\tau \in [0, H_p]$ and horizon $H_p \ge h$. A dot denotes differentiation with respect to physical time for actual signals and with respect to τ, holding t fixed, for predictions. The next update occurs at $t + h$. The symbol $\bar{R}_0$ denotes the externally supplied initial squared estimation-error bound. Time subscripts are omitted in pointwise derivations when no time dependence needs to be displayed. Parentheses denote function evaluation, such as $f_0(x_t, u_t)$.

C Well-posedness

Assumption 1 (Well-posed model and observer)**.** *The nominal model f_0 is continuously differentiable in state and input. Given initial states and admissible locally bounded measurable disturbance and control signals, the plant, observer, and nominal dynamics admit unique absolutely continuous solutions over the intervals considered.*

The certificate conditions and their operating domain are stated after the corresponding error inequalities are introduced.

III State-Dependent Estimation Error Bounds

A Recovering the global quadratic certificate

The starting point is the offline robust output-feedback design of Benders et al. [2]. Let $\epsilon \ge 0$ denote the constant estimation-error radius supplied by its global observer certificate, expressed here in the P-norm:

$$\|x - \hat{x}\|_P \le \epsilon, \qquad \text{equivalently} \qquad V_e \le \epsilon^2.$$

This invariant bound holds provided $V_{e,0} \le \epsilon^2$ and the certificate assumptions hold. We use the observer-certificate LMIs implemented in the accompanying ROHMPC repository [3], specialized here to a constant metric and written in the notation of this manuscript. This subsection recovers the original constant radius ϵ and introduces a time-varying squared radius R_t; Section III-B then constructs the local process-envelope reduction. For $b \in \mathbb{R}^n$, introduce the symmetric off-diagonal block

$$\mathcal{S}(b) = \begin{bmatrix} 0 & b \\ b^\top & 0 \end{bmatrix}. \tag{III.1}$$

Let $A_o = A - LC$ and $q_0 = \lambda\epsilon^2$, with $\lambda > 0$. Consider a baseline observer certificate with symmetric auxiliary matrices W, H satisfying

$$\mathcal{S}(E_0 w) \preceq W \qquad \forall w \in \mathcal{W}_0, \tag{III.2a}$$

$$\mathcal{S}(-LF\eta) \preceq H \qquad \forall \eta \in \mathcal{N}, \tag{III.2b}$$

$$W + H + \begin{bmatrix} XA_o^\top + A_o X + \lambda X & 0 \\ 0 & -q_0 \end{bmatrix} \preceq 0. \tag{III.2c}$$

Here $A = \partial f_0/\partial x$. The last inequality is a condition on the state Jacobians, and must hold along the segments joining the true and estimated states at their common input. Since X is constant, its time derivative is zero.

To see explicitly how these LMIs imply the error bound, first use the global baseline uncertainty $w_x \in \mathcal{W}_0$, writing $w = w_x$. Equation (II.4) can be written as

$$\dot{e} = A_o e + b, \qquad b := E_0 w - LF\eta.$$

For the nonlinear model, A here is the segment-averaged state Jacobian between $\hat{x}$ and x at the common input u. The following argument applies if the Jacobian LMI holds along that segment. The vector b collects the process disturbance and measurement-noise contributions.

Rewrite the first two inequalities in (III.2) as $\mathcal{S}(E_0 w) - W \preceq 0$ and $\mathcal{S}(-LF\eta) - H \preceq 0$, and add the third. The auxiliary matrices cancel:

$$\underbrace{\mathcal{S}(E_0 w) - W}_{\preceq 0} + \underbrace{\mathcal{S}(-LF\eta) - H}_{\preceq 0} + \underbrace{W + H + \operatorname{diag}(XA_o^\top + A_o X + \lambda X, -q_0)}_{\preceq 0} \preceq 0.$$

Since $\mathcal{S}$ is linear,

$$\mathcal{S}(E_0 w) + \mathcal{S}(-LF\eta) = \mathcal{S}(b),$$

so the resulting matrix inequality is

$$M := \begin{bmatrix} XA_o^\top + A_o X + \lambda X & b \\ b^\top & -q_0 \end{bmatrix} \preceq 0.$$

Negative semidefiniteness means $\xi^\top M \xi \le 0$ for every vector ξ. Choosing $\xi = [Pe; 1]$ and using $PX = XP = I$ and constant P gives

$$\begin{aligned} \begin{bmatrix} Pe \\ 1 \end{bmatrix}^\top M \begin{bmatrix} Pe \\ 1 \end{bmatrix} &= e^\top P(XA_o^\top + A_o X + \lambda X)Pe + 2e^\top Pb - q_0 \\ &= e^\top (A_o^\top P + PA_o)e + 2e^\top Pb + \lambda e^\top Pe - q_0 \\ &= \dot{V}_e + \lambda V_e - q_0 \le 0. \end{aligned}$$

Therefore,

$$\dot{V}_e \le -\lambda V_e + q_0. \tag{III.3}$$

Equivalently, define the invertible matrix $G = \operatorname{diag}(P, 1)$. The congruence transformation of M by G is

$$G^\top M G = \begin{bmatrix} A_o^\top P + PA_o + \lambda P & Pb \\ b^\top P & -q_0 \end{bmatrix} \preceq 0. \tag{III.4}$$

Indeed, $v^\top G^\top M G v = (Gv)^\top M (Gv) \le 0$ for every v; invertibility makes the two matrix inequalities equivalent. Evaluating the transformed matrix at $[e; 1]$ is identical to evaluating M at $[Pe; 1]$, because $G[e; 1] = [Pe; 1]$. These inequalities hold for every admissible w, η.

To bound the estimation error, introduce a scalar squared radius R_t defining the time-varying error set

$$\mathcal{E}_t = \{e : e^\top Pe \le R_t\}.$$

Thus R_t is the upper boundary in the scalar V_e coordinate, and $e^\top P e = R_t$ is the corresponding boundary in error-state space. Choose $R_0 \geq V_{e,0}$ and propagate this boundary using

$$\dot{R} = -\lambda R + q_0.$$

Subtracting this equation from (III.3) gives

$$\frac{d}{dt}(V_e - R) \leq -\lambda(V_e - R), \qquad V_{e,t} - R_t \leq e^{-\lambda t}(V_{e,0} - R_0) \leq 0.$$

Consequently, $V_{e,t} \leq R_t$ and $\sqrt{R_t}$ bounds the estimation error in the P-norm wherever the derivative inequality holds. The comparison equation has solution

$$R_t = \epsilon^2 + (R_0 - \epsilon^2)e^{-\lambda t}. \tag{III.5}$$

When $R_0 \geq V_{e,0}$ it bounds the estimation error; when $R_0 = \epsilon^2$, it exactly recovers the original constant radius ϵ.

B Local process-envelope reduction

The objective is to replace the global observer forcing q_0 by a smaller allowance that remains valid for the local process set. Starting from the model $d = E_0 w_x$, $w_x \in \mathcal{W}(x, u)$, we now specialize to a family parameterized by a scalar amplitude:

$$\mathcal{W}(x, u) = \mathcal{W}(x) = D(a(x))\mathcal{W}_0, \qquad \beta \leq a(x) \leq 1, \quad 0 < \beta < 1. \tag{III.6}$$

Thus $w_x = D(a(x))w$ for some $w \in \mathcal{W}_0$. Choose a fixed subset $\mathcal{J} \subseteq \{1, \ldots, m\}$ of disturbance components to scale, and define

$$D(a) = \operatorname{diag}(d_1(a), \ldots, d_m(a)), \qquad d_j(a) = \begin{cases} a, & j \in \mathcal{J}, \\ 1, & j \notin \mathcal{J}. \end{cases}$$

This construction is affine in a, has $D(1) = I$, and produces nested sets:

$$\beta \leq a_1 \leq a_2 \leq 1 \quad \Longrightarrow \quad E_0 D(a_1)\mathcal{W}_0 \subseteq E_0 D(a_2)\mathcal{W}_0. \tag{III.7}$$

The inclusion follows from the centered componentwise bounds, and remains valid after applying E_0.

The scalar structure serves two purposes. Affine dependence of $D(a)$ makes the vertex quadratics below convex in a, allowing a bound based on the two endpoint values. Nestedness permits an upper amplitude bound to enclose disturbances at all true states in a predicted tube. A general state-dependent matrix $D(x)$ can also define a disturbance model, but these properties need not hold: one would instead need to bound its process envelope over the whole tube, and the endpoint formula derived here would not follow automatically. Equation (III.6) specifies the structure used in this construction; the velocity law in Section III-C is one choice of $a(x)$. Let $W \in \mathbb{S}^{n+1}$ satisfy the baseline process-envelope inequality (III.2a), where $\mathbb{S}^{n+1}$ denotes the symmetric $(n+1) \times (n+1)$ matrices. Partition it as

$$W = \begin{bmatrix} Q & h_w \\ h_w^\top & c \end{bmatrix}, \qquad Q \succ 0. \tag{III.8}$$

Here $Q \in \mathbb{S}^n$ is the upper-left block, $h_w \in \mathbb{R}^n$ is the off-diagonal block, and $c \in \mathbb{R}$ is the bottom-right entry. These blocks belong to the given baseline certificate. The local construction assumes $Q \succ 0$, keeps Q and h_w fixed, and reduces the scalar entry c to accommodate smaller disturbance sets.

The Schur complement gives

$$\mathcal{S}(E_0 D(a) w) \preceq W \iff (E_0 D(a) w - h_w)^\top Q^{-1} (E_0 D(a) w - h_w) \le c. \tag{III.9}$$

The matrix Q^{-1} determines the quadratic weighting, h_w its center, and c the admissible level. The box $\mathcal{W}_0$ has 2^m vertices $w^{(i)}$, obtained by choosing each component as $\pm \bar{w}_j$. Define

$$\phi(w, a) := (E_0 D(a) w - h_w)^\top Q^{-1} (E_0 D(a) w - h_w), \qquad \phi_i(a) := \phi(w^{(i)}, a).$$

Here ϕ_i is the expression at the fixed vertex $w^{(i)}$, regarded as a function of the scale a. Their pointwise maximum is

$$g(a) = \max_{i=1,\dots,2^m} \phi_i(a) = \max_{w \in \mathrm{vert}(\mathcal{W}_0)} \phi(w, a). \tag{III.10}$$

For fixed a, the Hessian with respect to w is

$$\nabla_w^2 \phi(w, a) = 2 D(a)^\top E_0^\top Q^{-1} E_0 D(a) \succeq 0,$$

so $\phi(\cdot, a)$ is convex. Any $w \in \mathcal{W}_0$ can be written as $w = \sum_i \theta_i w^{(i)}$ with $\theta_i \ge 0$ and $\sum_i \theta_i = 1$. Therefore,

$$\phi(w, a) \le \sum_i \theta_i \phi(w^{(i)}, a) \le \max_i \phi_i(a) = g(a).$$

Since every box point is a convex combination of its vertices and the vertices belong to the box, $g(a) = \max_{w \in \mathcal{W}_0} \phi(w, a)$. Thus the vertex maximum bounds the entire disturbance box, including its interior.

For a fixed vertex, the affine dependence of $D(a)$ permits the representation

$$E_0 D(a) w^{(i)} - h_w = a p_i + r_i, \qquad \phi_i''(a) = 2 p_i^\top Q^{-1} p_i \ge 0,$$

where p_i, r_i are fixed vectors. Each vertex function is consequently convex in a, and their pointwise maximum g is convex as well, although it need not be differentiable where the maximizing vertex changes. Nestedness of $E_0 D(a) \mathcal{W}_0$ also gives $g(\beta) \le g(1)$.

A convex function lies below the chord joining its endpoint values. Writing $a = \frac{1-a}{1-\beta}\beta + \frac{a-\beta}{1-\beta} 1$ for $a \in [\beta, 1]$ yields

$$g(a) \le \frac{1-a}{1-\beta} g(\beta) + \frac{a-\beta}{1-\beta} g(1).$$

Equivalently,

$$g(a) \le g(1) - \frac{1-a}{1-\beta} (g(1) - g(\beta)). \tag{III.11}$$

For $\beta < a < 1$, the same inequality can be expressed as a comparison of secant slopes:

$$\frac{g(1) - g(a)}{1-a} \ge \frac{g(1) - g(\beta)}{1-\beta}.$$

The secant ending at 1 and starting at a has slope at least as large as the secant from β to 1. Hence the chord provides a guaranteed reduction from the global value $g(1)$, without requiring derivatives of g. The following proposition transfers this reduction to the process-envelope entry c and the observer forcing q_0 by reducing both by the same amount.

Proposition 1 (Local forcing from a fixed certificate)**.** *Suppose the baseline certificate* (III.2) *holds exactly, $Q \succ 0$, and the local disturbance sets have the structure* (III.6)*. Choose $0 \le \Delta \le g(1) - g(\beta)$ and define*

$$\theta(a) = \frac{1-a}{1-\beta}, \tag{III.12}$$

$$W(a) = \begin{bmatrix} Q & h_w \\ h_w^\top & c - \Delta \theta(a) \end{bmatrix}, \tag{III.13}$$

$$q(a) = q_0 - \Delta \theta(a). \tag{III.14}$$

For every $a \in [\beta, 1]$, the reduced envelope satisfies $\mathcal{S}(E_0 D(a) w) \preceq W(a)$ for all $w \in \mathcal{W}_0$. Together with the unchanged noise envelope, this gives the derivative bound $\dot{V}_e \leq -\lambda V_e + q(a)$.

Proof. At $a = 1$, $D(1) = I$. The process-envelope inequality (III.2a) of the global baseline certificate, together with the Schur-complement equivalence (III.9) and the vertex-maximum identity (III.10), implies

$$c \geq \max_{w \in \mathcal{W}_0} \phi(w, 1) = \max_i \phi_i(1) = g(1).$$

Equation (III.11) and the choice of Δ give $g(a) \leq c - \Delta\theta(a)$, proving the local envelope inequality by (III.9). To retain the dynamics certificate, reduce q_0 by the same amount as the bottom-right entry of W. Indeed, writing $B_o = X A_o^\top + A_o X + \lambda X$ gives

$$W(a) + H + \operatorname{diag}(B_o, -q(a)) = W + H + \operatorname{diag}(B_o, -q_0) \preceq 0.$$

Thus the disturbance envelope can shrink while the dynamics LMI remains identical. Adding the reduced process envelope, the unchanged noise envelope, and this dynamics inequality yields $\dot{V}_e \leq -\lambda V_e + q(a)$ by the quadratic-form calculation in Section III-A. □

At $a = 1$, $W(a) = W$ and $q(a) = q_0$. For smaller a, the chord bound gives a guaranteed process-envelope reduction, which is transferred to the observer forcing without changing the metric or gains. The construction therefore follows the sequence

$$\mathcal{W}(x) = D(a(x))\mathcal{W}_0 \;\longrightarrow\; g(a) \;\longrightarrow\; W(a) \;\longrightarrow\; q(a).$$

For use in MPC, x is unknown. We must next bound $a(x)$ over the predicted error tube so that the same local forcing is valid for every true state represented by that tube. For the positive-forcing formulation below, choose Δ such that $q_0 - \Delta > 0$.

C Velocity-Dependent Disturbance Bound over the Predicted Tube

Let S_v select the translational velocity components of the state, and denote the true velocity vector by $v_x := S_v x$. The symbol v in the nominal dynamics continues to denote the nominal control input. Consider the velocity-dependent amplitude law

$$a(x) = \beta + (1 - \beta)\frac{\|S_v x\|_2^2}{v_{\max}^2}, \qquad \|S_v x\|_2 \leq v_{\max}. \tag{III.15}$$

The amplitude enters the physical process disturbance through

$$d(x, w) = E_0 D(a(x)) w, \qquad w \in \mathcal{W}_0.$$

The matrix $D(a)$ is the component scaling defined in Section B. Thus a bound on true speed determines an enclosing local process-disturbance set.

Recall that $T^\top T = P$, so $\|e\|_P = \|Te\|_2$. Define

$$c_v := \left\| S_v T^{-1} \right\|_2. \tag{III.16}$$

Using the Cauchy–Schwarz inequality and the induced matrix norm bound, we obtain

$$\begin{aligned} \|S_v e\|_2 &= \left\| S_v T^{-1} (Te) \right\|_2 \\ &\leq \left\| S_v T^{-1} \right\|_2 \|Te\|_2 = c_v \|e\|_P \leq c_v \sqrt{R}. \end{aligned}$$

The same argument gives $\|S_v\delta\|_2 \le c_v \|\delta\|_P \le c_v s$. Starting from the state decomposition $x = z + \delta + e$, the triangle inequality gives

$$\begin{aligned}\|S_v x\|_2 &= \|S_v z + S_v\delta + S_v e\|_2 \\ &\le \|S_v z\|_2 + \|S_v\delta\|_2 + \|S_v e\|_2 \\ &\le \|S_v z\|_2 + c_v(s + \sqrt{R}).\end{aligned}$$

Combining this tube bound with the certified domain bound $\|S_v x\|_2 \le v_{\max}$ defines

$$v_{\text{up}} := \min\{v_{\max}, \|S_v z\|_2 + c_v(\sqrt{R} + s)\}. \tag{III.17}$$

Thus v_{up} bounds true speed for every state in the predicted error tube that lies in the certified domain. The cap $v_{\max}$ requires domain containment of the true-state tube; a bound on the nominal speed alone does not establish it.

Since the amplitude law (III.15) is nondecreasing in speed,

$$a(x) \le a_{\text{up}} := \beta + (1-\beta)(v_{\text{up}}/v_{\max})^2. \tag{III.18}$$

The set inclusion (III.7) then gives

$$E_0 D(a(x))\mathcal{W}_0 \subseteq E_0 D(a_{\text{up}})\mathcal{W}_0.$$

Applying Proposition 1 to this enclosing set yields

$$\dot{V}_e \le -\lambda V_e + q(a_{\text{up}}), \qquad q(z, R, s) := q(a_{\text{up}}). \tag{III.19}$$

More explicitly, the scalar used in the estimation-radius dynamics is

$$q(z, R, s) = q_0 - \Delta \frac{1 - a_{\text{up}}(z, R, s)}{1 - \beta}.$$

The local uncertainty model specifies $a(x)$, while the predicted state and radii determine its enclosing value a_{up}. The resulting $q(z, R, s)$ is therefore a bound valid throughout that tube. Here q has no explicit dependence on the nominal input v. Section IV writes $q(z, v, R, s)$ for the general case.

D Comparison with Separate-Norm Disturbance Bounds

Retaining the model, metric, and gains from Section II and the uncertainty sets from Section III-B, consider an alternative estimation-error bound obtained by bounding the process-disturbance and measurement-noise contributions separately. Derivative inequalities for error norms are understood to hold almost everywhere in time. If the disturbance-free observer error satisfies $\frac{d}{dt}\|e\|_P \le -\rho_o \|e\|_P$ for a contraction rate $\rho_o > 0$, a radius bound can be propagated as

$$\dot{r} = -\rho_o r + b_{\text{rest}} + a_{\text{up}} b_{\text{acc}} + b_\eta. \tag{III.20}$$

Here $b_{\text{acc}} = \max \|TE_{\text{acc}} w_{\text{acc}}\|_2$ bounds the unscaled acceleration-disturbance contribution, b_{rest} bounds the remaining process contribution, and $b_\eta = \max_{\eta\in\mathcal{N}} \|TLF\eta\|_2$ bounds measurement noise. Each maximum is taken over its corresponding admissible set.

For vector contributions $d_j(\omega_j)$ with $\omega_j \in \Omega_j$, this construction uses

$$\max_{\omega \in \prod_j \Omega_j} \left\| \sum_j d_j(\omega_j) \right\|_P \le \sum_j \max_{\omega_j \in \Omega_j} \|d_j(\omega_j)\|_P .$$

The triangle inequality and separate maximizations discard directional information: the individual largest contributions need not align to attain the sum of their norms. Thus this approach can be more conservative and need not recover the original SDP radius at $a = 1$. Proposition 1 retains the original quadratic certificate and recovers its global baseline at $a = 1$.

IV Robust Output-Feedback MPC

A Coupled tube dynamics

Suppose the tracking contraction condition holds along every state–input segment joining (z, v) to $(\hat{x}, v + K(\hat{x} - z))$ in the certified domain:

$$(A + BK)^\top P + P(A + BK) \preceq -2\rho P, \qquad \rho > 0. \tag{IV.1}$$

With $T^\top T = P$, define

$$\gamma := \left\| TLCT^{-1} \right\|_2, \qquad b_\eta := \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2 .$$

The constant b_η bounds the measurement-noise contribution $LF\eta$ to the observer dynamics in the P-norm. Equation (II.4) then implies

$$\frac{d}{dt} \|\delta\|_P \le -\rho \|\delta\|_P + \gamma \sqrt{V_e} + b_\eta. \tag{IV.2}$$

The derivation of (IV.2), using the contraction condition (IV.1) and the Cauchy–Schwarz inequality, is given in Appendix A. Thus the proposed augmented prediction is

$$\dot{z} = f_0(z, v), \tag{IV.3a}$$
$$\dot{R} = -\lambda R + q(z, v, R, s), \tag{IV.3b}$$
$$\dot{s} = -\rho s + \gamma \sqrt{R} + b_\eta. \tag{IV.3c}$$

The general notation permits input dependence in q; the present velocity example has none explicitly. The forcing bound $q(z, v, R, s)$ must be valid for every state–input pair represented by the predicted tube and every admissible disturbance realization.

Assumption 2 (Certificate validity on the operating domain)**.** *Let $\mathcal{D} \subseteq \mathbb{R}^n \times \mathbb{R}^{n_u}$ be a specified state–input region. The baseline observer LMIs* (III.2) *and tracking contraction condition* (IV.1) *hold on a neighborhood of $\mathcal{D}$. The disturbance-set model is valid there. For any predicted (z, v, R, s) and errors satisfying $e^\top P e \le R$ and $\|\delta\|_P \le s$, set $\hat{x} = z + \delta$, $x = z + \delta + e$, and $u = v + K\delta$. The certificate applies whenever the segments*

$$\{(\hat{x} + \theta e, u) : 0 \le \theta \le 1\},$$
$$\{(z + \theta\delta, v + \theta K\delta) : 0 \le \theta \le 1\}$$

lie in $\mathcal{D}$. The forcing $q(z, v, R, s)$ encloses the process uncertainty for every such true state and input, as constructed in Section III-C for the velocity-dependent example. The resulting error inequalities are (III.19) *and* (IV.2)*.*

The two segments connect true and estimated states at a common input, and nominal and estimated states with the feedback input, respectively. To keep these certificates applicable, the MPC imposes the domain-containment condition that both segments lie in $\mathcal{D}$ for every error pair in its predicted tube. This is a constraint on the prediction, distinct from the certificate inequalities and from the physical state–input constraints. Finite Jacobian-grid checks alone do not establish the domain-wide inequalities in Assumption 2.

Assumption 3 (Radius regularity and initial bounds)**.** *The forcing bound $q(z, v, R, s)$ is locally Lipschitz in its arguments on the set of admissible nominal states, inputs, and radius pairs, and satisfies $0 < q_{\min} \le q \le q_{\max}$. Define*

$$0 < R_{\min} \le q_{\min}/\lambda, \quad R_{\max} = q_{\max}/\lambda, \quad s_{\max} = (\gamma \sqrt{R_{\max}} + b_\eta)/\rho. \tag{IV.4}$$

The pair of radius variables (R, s) is constrained to the rectangle

$$\begin{aligned}\mathcal{B} &= [R_{\min}, R_{\max}] \times [0, s_{\max}] \\ &= \{(R, s) : R_{\min} \le R \le R_{\max},\ 0 \le s \le s_{\max}\}.\end{aligned}$$

These variables bound the errors through $e^\top P e \le R$ and $\|\delta\|_P \le s$. An available initial prior $\bar{R}_0 \in [R_{\min}, R_{\max}]$ satisfies $V_{e,0} \le \bar{R}_0$. The observer state evolves continuously according to the observer dynamics in Section II, including across MPC updates.

The coupled radius equations (IV.3) satisfy the following conditions on the four boundaries of $\mathcal{B}$:

$$\begin{aligned} R = R_{\min}: &\quad \dot{R} \ge -\lambda R_{\min} + q_{\min} \ge 0, \\ R = R_{\max}: &\quad \dot{R} \le -\lambda R_{\max} + q_{\max} = 0, \\ s = 0: &\quad \dot{s} = \gamma\sqrt{R} + b_\eta \ge 0, \\ s = s_{\max}: &\quad \dot{s} \le -\rho s_{\max} + \gamma\sqrt{R_{\max}} + b_\eta = 0. \end{aligned}$$

Thus the radius pair cannot leave this rectangle along admissible nominal trajectories: if $(R_0, s_0) \in \mathcal{B}$, then $(R_t, s_t) \in \mathcal{B}$ for all subsequent times on which the assumptions hold. Since $\frac{d}{dR}\sqrt{R} = \frac{1}{2\sqrt{R}}$, the derivative is unbounded as $R \to 0^+$. Restricting $R \ge R_{\min} > 0$ bounds this derivative and ensures that the square-root term is Lipschitz on the admissible interval.

Lemma 1 (Tube containment)**.** *Under Assumptions 1–3, the observer and tracking derivative inequalities, and valid initial bounds, $V_{e,t} \le R_t$ and $\|\delta_t\|_P \le s_t$ throughout an applied prediction interval.*

Proof. On $V_e = R$, the derivative of $V_e - R$ is nonpositive. Subtracting (IV.3c) from (IV.2) at $\|\delta\|_P = s$, with $V_e \le R$, gives $\frac{d}{dt}(\|\delta\|_P - s) \le \gamma(\sqrt{V_e} - \sqrt{R}) \le 0$. These boundary inequalities, together with the regularity assumptions, ensure that the estimation and tracking errors remain within their coupled bounds. Domain inclusion and a first-exit continuation argument ensure that the derivative hypotheses remain applicable. □

For identical nominal trajectories and initial radii, the local estimation-error forcing bound $q_{\text{local,t}} \le q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound, yields $R_{\text{local,t}} \le R_{\text{global,t}}$ and $s_{\text{local,t}} \le s_{\text{global,t}}$. Thus the local formulation produces no larger predicted error bounds under these common conditions. The derivation is given in Appendix B.

B Tightened constraints and MPC initialization

For each state–input constraint $g_j(x, u) \le 0$, impose the worst-case condition over the predicted error tube:

$$\sup_{\|e\|_P \le \sqrt{R},\ \|\delta\|_P \le s} g_j(z + \delta + e, v + K\delta) \le 0. \tag{IV.5}$$

The support function of a set $\mathcal{E}$ is

$$h_{\mathcal{E}}(a) := \sup_{\xi \in \mathcal{E}} a^\top \xi,$$

which gives its largest projection in direction a. For an affine constraint $a_j^\top x + b_j^\top u \le c_j$, substituting $x = z + \delta + e$ and $u = v + K\delta$ gives the uncertain contribution $(a_j^\top + b_j^\top K)\delta + a_j^\top e$. Evaluating the support functions over the two error balls, with $T^\top T = P$, defines the tightening constants

$$c_{s,j} := \left\|(a_j^\top + b_j^\top K)T^{-1}\right\|_2, \qquad c_{e,j} := \left\|a_j^\top T^{-1}\right\|_2.$$

The norms are Euclidean norms of the indicated row vectors. For fixed affine constraints, metric P, and gain K, these constants can be computed offline. The resulting tightened constraint is

$$a_j^\top z + b_j^\top v + \underbrace{c_{s,j} s}_{\text{tracking-error margin}} + \underbrace{c_{e,j}\sqrt{R}}_{\text{estimation-error margin}} \le c_j. \tag{IV.6}$$

This expression evaluates the supremum exactly over the chosen product of estimation and tracking error balls. The margins vary along the prediction through R and s. Using the time notation in Section II, the tightened state–input constraints (IV.5) (or (IV.6) for affine constraints), certificate-domain conditions in Assumption 2, and radius bounds $(R_{\tau|t}, s_{\tau|t}) \in \mathcal{B}$ are imposed for every $\tau \in [0, H_p]$. At update t, minimize the predicted horizon cost

$$J_t = \int_0^{H_p} \ell(z_{\tau|t}, v_{\tau|t})\, d\tau + V_f(z_{H_p|t}), \tag{IV.7}$$

where ℓ is the stage-cost rate, its integral is the accumulated running cost, and V_f is the terminal cost. Initialize $R_{0|0} = \bar{R}_0$ using Assumption 3. At subsequent updates, retain the previous plan's propagated bound:

$$R_{0|t+h} = R_{h|t}. \tag{IV.8}$$

The prediction satisfies (IV.3), the constraints above, and

$$\left\|\hat{x}_t - z_{0|t}\right\|_P \le s_{0|t}, \quad (z_{H_p|t}, R_{H_p|t}, s_{H_p|t}) \in \Omega_f. \tag{IV.9}$$

The initial nominal center $z_{0|t}$ is free. For the selected feasible plan, apply

$$u_{t+\tau} = v_{\tau|t} + K(\hat{x}_{t+\tau} - z_{\tau|t}), \qquad 0 \le \tau < h,$$

and initialize the next prediction with $R_{0|t+h} = R_{h|t}$. Lemma 1 ensures $V_{e,t+h} \le R_{h|t}$, so this remains a valid estimation-error bound. Starting from an initially feasible MPC problem, recursive feasibility and constraint satisfaction hold when the controller applies a feasible plan at each update; optimality is not required for these guarantees. Superscript $*$ denotes optimal plans and the minimum horizon cost in the cost-decrease inequality (IV.11) of Theorem 1.

C Terminal Ingredients and Recursive Feasibility

Assumption 4 (Terminal ingredients)**.** *There exist a nonempty nominal terminal set Z_f and controller κ_f such that $\dot{z} = f_0(z, \kappa_f(z))$ leaves Z_f invariant. For every $z \in Z_f$, with nominal input $v = \kappa_f(z)$, the tightened state–input constraints* (IV.5) *and the certificate-domain conditions in Assumption 2 hold at $R = R_{\max}$ and $s = s_{\max}$. For the cost assertion additionally assume $\ell, V_f \ge 0$ and*

$$\nabla V_f(z) f_0(z, \kappa_f(z)) \le -\ell(z, \kappa_f(z)). \tag{IV.10}$$

Then $\Omega_f = Z_f \times \mathcal{B}$ is invariant under terminal control and the radius dynamics. The terminal set is certified using the largest admissible estimation and tracking radii. Since the smaller error balls are contained in these largest ones, this ensures terminal constraint satisfaction and certificate-domain containment for every $(R, s) \in \mathcal{B}$, while the predictions along the horizon retain their local, time-varying radii.

Theorem 1 (Recursive Feasibility and Robust Constraint Satisfaction)**.** *Suppose Assumptions 1–4 and the estimation-error inequality* (III.19)*, written as $\dot{V}_e \le -\lambda V_e + q(z, v, R, s)$, and tracking-error inequality* (IV.2) *hold exactly, and the initial MPC problem is feasible. Applying a feasible plan satisfying* (IV.9) *and initializing $R_{0|0} = \bar{R}_0$ and $R_{0|t+h} = R_{h|t}$ at subsequent updates gives recursive feasibility and continuous-time robust constraint satisfaction for all disturbances in the assumed sets. If* (IV.10) *holds and the selected feasible plans attain the minimum horizon cost among all feasible plans, then*

$$J^*_{t+h} \le J^*_t - \int_0^h \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau. \tag{IV.11}$$

Proof. **Part 1: Recursive feasibility.** Suppose the plan $(z_{\tau|t}, v_{\tau|t}, R_{\tau|t}, s_{\tau|t})$ is feasible at update t. For $0 \le \tau \le H_p - h$, define

$$(z^c_{\tau|t+h}, v^c_{\tau|t+h}, R^c_{\tau|t+h}, s^c_{\tau|t+h}) = (z_{\tau+h|t}, v_{\tau+h|t}, R_{\tau+h|t}, s_{\tau+h|t}).$$

Evaluating the shifted-candidate definition at $\tau = 0$ gives

$$z^c_{0|t+h} = z_{h|t}, \qquad R^c_{0|t+h} = R_{h|t}, \qquad s^c_{0|t+h} = s_{h|t}.$$

The estimation-radius update rule (IV.8) sets $R_{0|t+h} = R_{h|t}$, so the candidate has the required initial radius $R^c_{0|t+h} = R_{0|t+h}$. Lemma 1, applied over the previous control interval, ensures $V_{e,t+h} \le R_{h|t}$ and

$$\left\|\hat{x}_{t+h} - z_{h|t}\right\|_P \le s_{h|t}.$$

Using the shifted initial center and tracking radius therefore yields

$$\left\|\hat{x}_{t+h} - z^c_{0|t+h}\right\|_P = \left\|\hat{x}_{t+h} - z_{h|t}\right\|_P \le s_{h|t} = s^c_{0|t+h}.$$

Thus the candidate satisfies the prescribed estimation-radius initialization and the initial tracking bound in (IV.9). On this shifted interval the candidate reproduces the previous feasible trajectory. The coupled nominal and tube dynamics (IV.3) and prediction constraints have no explicit time dependence. Consequently, the shifted candidate continues to satisfy those dynamics, the tightened constraints (IV.5), certificate-domain conditions, and radius bounds.

At $\tau = H_p - h$, the candidate reaches the previous terminal point in Ω_f. Over the remaining h units of time, apply $v^c_{\tau|t+h} = \kappa_f(z^c_{\tau|t+h})$ and continue the nominal and radius dynamics. Terminal invariance under Assumption 4 and invariance of $\mathcal{B}$ keep this continuation in Ω_f, where all tightened and domain constraints hold. Its endpoint therefore satisfies the new terminal condition. This constructs a feasible plan at update $t + h$; induction from initial feasibility proves recursive feasibility.

Part 2: Robust constraint satisfaction. During each applied interval, Lemma 1 gives $\|e\|_P \le \sqrt{R_{\tau|t}}$ and $\|\delta\|_P \le s_{\tau|t}$. The actual pair satisfies $x_{t+\tau} = z_{\tau|t} + \delta + e$ and $u_{t+\tau} = v_{\tau|t} + K\delta$, and hence belongs to the set over which (IV.5) takes its supremum. Consequently,

$$g_j(x_{t+\tau}, u_{t+\tau}) \le \sup_{\|e\|_P \le \sqrt{R_{\tau|t}},\ \|\delta\|_P \le s_{\tau|t}} g_j(z_{\tau|t} + \delta + e, v_{\tau|t} + K\delta) \le 0.$$

This holds throughout every applied interval and for all admissible disturbances.

Part 3: Optimal-cost decrease. Now take an optimal plan at update t. Let $z_{f,\sigma}$, $0 \le \sigma \le h$, be the appended terminal trajectory, with $z_{f,0} = z^*_{H_p|t}$ and input $\kappa_f(z_{f,\sigma})$. The shifted candidate has cost

$$\begin{aligned} J^c_{t+h} = &\int_h^{H_p} \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau \\ &+ \int_0^h \ell(z_{f,\sigma}, \kappa_f(z_{f,\sigma}))\, d\sigma + V_f(z_{f,h}). \end{aligned}$$

Integrating (IV.10) along the terminal continuation gives

$$\int_0^h \ell(z_{f,\sigma}, \kappa_f(z_{f,\sigma}))\, d\sigma + V_f(z_{f,h}) \le V_f(z^*_{H_p|t}).$$

The terminal-cost decrease therefore covers the accumulated stage cost of the appended interval. Substituting this bound and comparing with (IV.7) yields

$$\begin{aligned} J^c_{t+h} &\le \int_h^{H_p} \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau + V_f(z^*_{H_p|t}) \\ &= J^*_t - \int_0^h \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau. \end{aligned}$$

Since the next optimal cost is no greater than the cost of this feasible candidate, $J^*_{t+h} \le J^c_{t+h}$, which establishes (IV.11). □

The shifted-and-extended candidate provides a feasible fallback. If a suboptimal solver returns a feasible plan whose horizon cost does not exceed the candidate cost, the same argument establishes the cost-decrease inequality for the selected plans.

Remark 1 (Relation to [1]). Our formulation follows the free-center approach in Section IV-B of [1]: the optimizer selects the initial nominal center subject to the tracking-error bound. The shifted center $z^c_{0|t+h} = z_{h|t}$ belongs to the feasible candidate constructed in the proof of Theorem 1; it is not imposed on the newly optimized plan. The estimation radius is initialized using (IV.8). Section IV-C of [1] instead fixes the initial nominal center to the current estimate and establishes recursive feasibility under its corresponding terminal conditions. Here, we retain the free-center architecture and incorporate the state-dependent forcing derived in Section III.

The horizon-cost decrease in (IV.11) bounds the accumulated nominal running cost. By (II.3), the true-state deviation satisfies

$$\|x - x_{\mathrm{ref}}\|_P \le \|z - x_{\mathrm{ref}}\|_P + \sqrt{R} + s.$$

Consequently, if the applied nominal trajectory converges to x_{ref}, then

$$\limsup_{t\to\infty} \|x_t - x_{\mathrm{ref}}\|_P \le \sqrt{R_{\max}} + s_{\max},$$

where t denotes continuous time. Nominal convergence is an additional condition here; it is not established by the cost-decrease inequality alone.

V Numerical Example: Falcon Model

A Model and reproduced offline design

The example uses the repository [3] and the saved Falcon design from 10 September 2026. The 12-state model consists of position, attitude, translational velocity, and body angular velocity, with four rotor inputs. The nominal translational drag matrix is zero in this instance. Its Jacobians are therefore independent of translational velocity; this does not restrict the controller to hovering. The velocity domain is $[-2, 2]^3$ m/s, giving $v_{\max} = \sqrt{12}$ m/s.

The original 1024-point Jacobian grid varies four rotor inputs, three angles, and three body rates at two endpoints each. Positions and translational velocities are zero at these points. This grid alone does not certify the continuous domain of the remaining nonlinear coordinates.

The reproduced robust-design and terminal-cost SDPs returned `OPTIMAL`, with objectives 1494.858118 and 38.564610, respectively. The first is a weighted combination of normalized squared tightening coefficients and squared estimation radius, not an optimized contraction rate. The second minimizes the trace of the terminal-cost matrix. Solver success does not ensure useful tightened constraints: the reproduced motor-input half-width was approximately 0.1200, whereas the associated global margin was 0.4619, leaving negative available width. Consequently, this design has not supplied the nonempty terminal set required by Theorem 1.

B Prescribed local uncertainty and parameters

For the Falcon state ordering (position, attitude, translational velocity, body angular velocity), the velocity selector and disturbance scaling matrix are

$$S_v = \begin{bmatrix} 0_{3\times 6} & I_3 & 0_{3\times 3} \end{bmatrix},$$
$$D(a) = \mathrm{diag}(I_6, aI_3, I_3).$$

The scaled disturbance components enter the translational acceleration equations. Only the centered acceleration disturbance components 7–9 are scaled. Other process-disturbance components, measurement noise, and the nominal bias remain unchanged. We use $\beta = 0.25$ and the velocity law in Section III-C. This is a synthetic uncertainty model, not a new identification or UQ result. The experiment reuses the saved $X, P, L, K, W, H, \lambda$ and solves no new SDP. The factor 0.99

Table 1: Saved certificate and local-envelope parameters (rounded).

Quantity	Value
Observer decay coefficient λ	0.497029
Original uncalibrated ϵ	0.07442638619
Global forcing q_0	0.002753187883
$g(1)$	0.00148773
$g(\beta)$	0.00104178
$\Delta = 0.99[g(1) - g(\beta)]$	0.000441487
$q(\beta)$	0.0023117
Condition number of Q	3474.67

retains a numerical cushion but does not repair an infeasible baseline certificate or provide a rigorous floating-point error bound.

C Radius propagation experiment

A prescribed nominal speed is 0 m/s on $[0, 8)$ s, 2.5 m/s on $[8, 16)$ s, and 0 m/s on $[16, 32]$ s. Initialize $R_0 = \epsilon^2$, $s_0 = 0$. Both cases use identical metrics, gains, initial bounds, tracking dynamics, and noise; only the local forcing versus q_0 changes. The global estimation radius equals ϵ by (III.5). The final estimation-radius reduction is $100(1 - r_{\text{local}}/\epsilon) = 6.68019\%$. This is

Table 2: Final-time comparison at 32 s; these are bounds, not measured errors.

Quantity	Global	State-dependent
Estimation radius r	0.07442638619	0.06945456129
Tracking radius s	0.392384	0.367532

neither a time-average improvement nor a universal performance guarantee. The speed schedule is not a dynamically feasible quadrotor trajectory, and no MPC optimization or closed-loop plant simulation is performed. The earlier separate-norm prototype produced a global equilibrium radius of approximately 0.105718 and a final local radius of 0.0979251, both above the original 0.0744264. These values belong to a different bound construction and do not indicate empirical calibration of the original ϵ. The quadratic-envelope approach restores a baseline-equivalent comparison.

D Numerical checks and limitations

A numerical MPC implementation that checks constraints only at selected prediction times must additionally establish constraint satisfaction between those times. The continuous-time formulation in Section IV-B imposes them throughout the horizon. The current experiment checks all 4096 original process vertices, all noise vertices, the 1024 original dynamics points, and all process vertices at 31 scale values. It additionally evaluates 1968 admissible random nonlinear error configurations. Table 3 reports the saved residuals. The saved feasibility tolerance is 10^{-6}. Small positive residuals mean feasibility within that numerical tolerance, not exact semidefinite feasibility. The convex-chord argument covers intermediate scales analytically only conditional

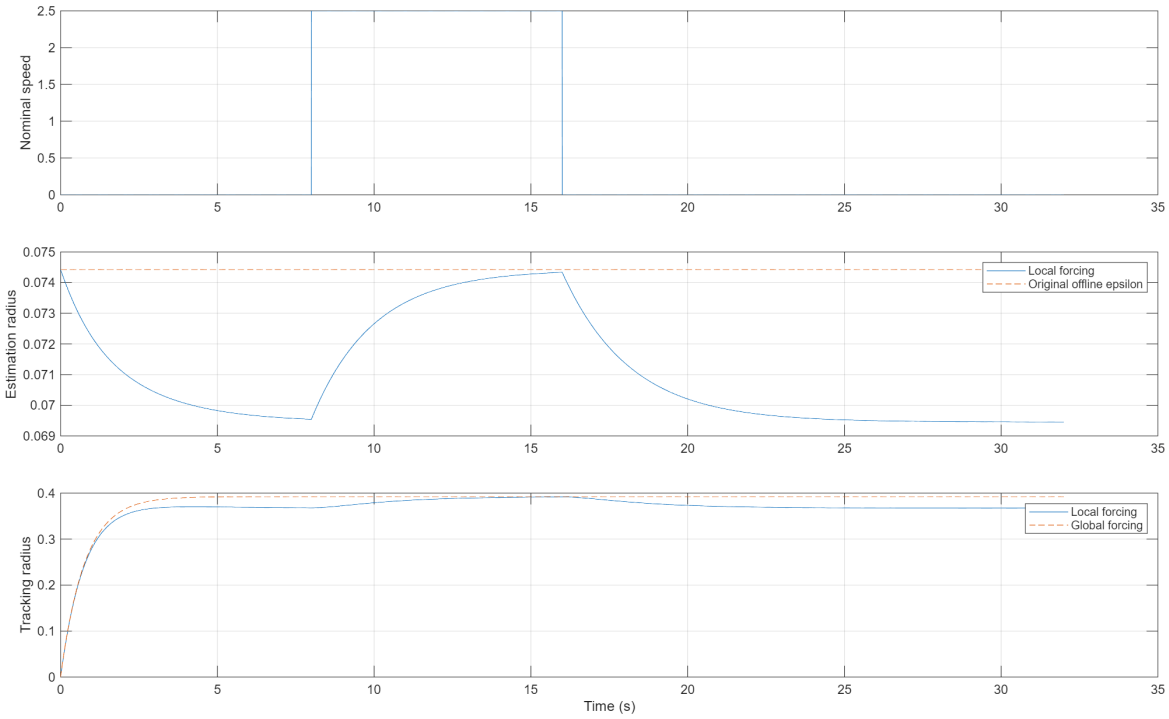


Figure 1: Recorded tube-ODE experiment with a prescribed low/high/low nominal-speed schedule. The figure is copied from the saved run, not generated by a closed-loop simulator.

Table 3: Numerical audit; positive matrix residuals are tolerated numerically, not exact certificates.

Check	Maximum residual
Original process-envelope LMI	$+5.50832\,10^{-8}$
Original noise-envelope LMI	$+5.57035\,10^{-8}$
Observer dynamics LMI	$+5.45468\,10^{-8}$
Tracking LMI	-0.0030671
Local Schur envelope	$+5.63813\,10^{-8}$
Sampled nonlinear derivative excess	-0.00229595

on exact baseline inequalities. Finite Jacobian and nonlinear samples do not prove full-domain coverage. No closed-loop feasibility, safety, or stability claim for Falcon follows from these tests.

VI Conclusion

A fixed observer certificate can support locally reduced process forcing through a Schur-complement envelope construction. Coupling this forcing to a tracking radius yields a candidate state-dependent output-feedback MPC formulation with conditional recursive-feasibility and safety guarantees. The Falcon radius experiment demonstrates a smaller local bound relative to a baseline-equivalent global propagation. Remaining work is to certify the continuous domain and numerical margins, validate the local uncertainty model, construct compatible nonempty terminal ingredients, and then implement and evaluate the closed-loop controller.

References

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[2] D. Benders, J. Köhler, R. Babuška, J. Alonso-Mora, and L. Ferranti, "From Data to Safe Mobile Robot Navigation: An Efficient and Modular Robust MPC Design Pipeline," arXiv:2508.07045v1, 2025. `https://arxiv.org/abs/2508.07045`.

[3] D. Benders et al., ROHMPC code repository, `https://github.com/dbenders1/rohmpc`. Numerical inputs and saved outputs from the local reproduction of 10 September 2026.

APPENDIX

A Derivation of the Tracking-Error Bound

Recall $\delta = \hat{x} - z$, $e = x - \hat{x}$, and $u = v + K\delta$. Subtracting the nominal dynamics from the observer dynamics gives

$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \tag{A.1}$$

For fixed z, v, δ, parameterize the straight-line state–input segment by

$$x_\theta = z + \theta\delta, \qquad u_\theta = v + \theta K\delta, \qquad 0 \le \theta \le 1.$$

Here θ parameterizes a segment in state–input space, not time. Its endpoints are (z, v) and $(\hat{x}, u)$. With $A = \partial f_0/\partial x$ and $B = \partial f_0/\partial u$, integration along this segment yields

$$f_0(\hat{x}, u) - f_0(z, v) = \int_0^1 \big[A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K\big]\,\delta\, d\theta \tag{A.2}$$

$$= \bar{A}_K\delta, \qquad \bar{A}_K := \int_0^1 \big[A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K\big]d\theta. \tag{A.3}$$

The domain assumptions ensure the contraction inequality holds at every point of this segment. Integrating it and using constant P gives

$$\bar{A}_K^\top P + P\bar{A}_K \preceq -2\rho P. \tag{A.4}$$

Define $d_o := LCe + LF\eta$. Then (A.1) becomes $\dot{\delta} = \bar{A}_K\delta + d_o$. For $\ell_\delta := \|\delta\|_P = \sqrt{\delta^\top P\delta} > 0$,

$$\dot{\ell}_\delta = \frac{\delta^\top P\dot{\delta}}{\ell_\delta} \tag{A.5}$$

$$= \frac{\delta^\top P\bar{A}_K\delta}{\ell_\delta} + \frac{\delta^\top P d_o}{\ell_\delta}. \tag{A.6}$$

The contraction contribution satisfies

$$\delta^\top P\bar{A}_K\delta = \frac{1}{2}\delta^\top(P\bar{A}_K + \bar{A}_K^\top P)\delta \le -\rho\ell_\delta^2.$$

For the forcing contribution, $T^\top T = P$ and the Cauchy–Schwarz inequality imply

$$\frac{\delta^\top P d_o}{\ell_\delta} = \frac{(T\delta)^\top(Td_o)}{\|T\delta\|_2} \le \|Td_o\|_2 .$$

The triangle inequality and the definitions of γ and b_η give

$$\|Td_o\|_2 \le \|TLCe\|_2 + \|TLF\eta\|_2 \tag{A.7}$$

$$\le \left\|TLCT^{-1}\right\|_2 \|Te\|_2 + \max_{\eta\in\mathcal{N}} \|TLF\eta\|_2 \tag{A.8}$$

$$= \gamma\sqrt{V_e} + b_\eta. \tag{A.9}$$

Combining these estimates establishes

$$\frac{d}{dt}\|\delta\|_P \le -\rho\|\delta\|_P + \gamma\sqrt{V_e} + b_\eta, \tag{A.10}$$

which is (IV.2). The calculation involving division by ℓ_δ applies when $\delta \neq 0$. Along absolutely continuous trajectories, the norm is absolutely continuous and its derivative is zero almost

everywhere on the set where the norm vanishes. Since the forcing bound is nonnegative, the same inequality therefore holds almost everywhere in time, including that zero set.

Finally, $V_e \le R$ gives the comparison dynamics

$$\dot{s} = -\rho s + \gamma\sqrt{R} + b_\eta, \qquad s_0 \ge \|\delta_0\|_P .$$

Subtracting this equation from the norm inequality and applying an integrating factor yields

$$\|\delta_t\|_P - s_t \le e^{-\rho t}\big(\|\delta_0\|_P - s_0\big) \le 0,$$

provided the estimation bound and certificate-domain conditions remain valid, as required in the coupled containment argument.

B Comparison of Local and Global Radius Bounds

Consider the same nominal state and input trajectories z_t, v_t, identical initial radius values, and the same metric, gains, and constants $\lambda, \rho, \gamma, b_\eta$ in both propagations. Define the local estimation-error forcing along its radius trajectory by

$$q_{\mathrm{local,t}} := q\big(z_t, v_t, R_{\mathrm{local,t}}, s_{\mathrm{local,t}}\big).$$

For the envelope construction in Proposition 1, $q_{\mathrm{local,t}} \le q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound. The comparison below holds on any interval on which both propagations are well-defined and their domain assumptions hold.

The squared estimation radii satisfy

$$\dot{R}_{\mathrm{local}} = -\lambda R_{\mathrm{local}} + q_{\mathrm{local,t}}, \tag{B.1}$$

$$\dot{R}_{\mathrm{global}} = -\lambda R_{\mathrm{global}} + q_0, \qquad R_{\mathrm{local},0} = R_{\mathrm{global},0}. \tag{B.2}$$

Let $\Delta_R = R_{\mathrm{local}} - R_{\mathrm{global}}$. Subtracting the equations and multiplying by $e^{\lambda t}$ gives

$$\frac{d}{dt}(e^{\lambda t}\Delta_{R,t}) = e^{\lambda t}(q_{\mathrm{local,t}} - q_0).$$

Integration, using $\Delta_{R,0} = 0$, yields

$$R_{\mathrm{local,t}} - R_{\mathrm{global,t}} = \int_0^t e^{-\lambda(t-\tau)}(q_{\mathrm{local},\tau} - q_0)\, d\tau \le 0. \tag{B.3}$$

The inequality follows because the exponential is positive and the forcing difference is nonpositive. This argument permits q_{local} to depend on the local radius trajectory; it only uses its upper bound by q_0.

The corresponding tracking radii satisfy

$$\dot{s}_{\mathrm{local}} = -\rho s_{\mathrm{local}} + \gamma\sqrt{R_{\mathrm{local}}} + b_\eta, \tag{B.4}$$

$$\dot{s}_{\mathrm{global}} = -\rho s_{\mathrm{global}} + \gamma\sqrt{R_{\mathrm{global}}} + b_\eta, \qquad s_{\mathrm{local},0} = s_{\mathrm{global},0}. \tag{B.5}$$

Subtracting and applying the integrating factor $e^{\rho t}$ similarly gives

$$\begin{aligned} s_{\mathrm{local,t}} - s_{\mathrm{global,t}} &= \gamma \int_0^t e^{-\rho(t-\tau)} \left(\sqrt{R_{\mathrm{local},\tau}} - \sqrt{R_{\mathrm{global},\tau}}\right) d\tau \\ &\le 0. \end{aligned} \tag{B.6}$$

Here (B.3), nonnegative radii, and $\gamma \ge 0$ establish the final inequality. Consequently, under the common nominal trajectories and initialization, both predicted error bounds in the local formulation are no larger than their global baseline counterparts.